MASTER COURSE BRIEF: pH RAINBOW

A COLORFUL CHEMISTRY ADVENTURE WITH MAGIC JUICE!

1. Document Control & Course Overview

Course ID	PLANRS-SCI-GRK2-004
Target Grade Level	Kindergarten to Grade 2 (Ages 5–8)
Subject Area	Foundational Chemistry & Experimental Sciences
Core Topic	Acids, Bases, and Natural Indicators (pH Spectrum)
Estimated Duration	60 to 90 Minutes (Adaptive Live Session)
Developer Institution	Planrs Academy Curriculum Engineering Division
Publication Date	June 2026

Course Narrative Summary

The "pH Rainbow" module is designed as an interactive, narrative-driven chemistry workshop optimized for early childhood education. By framing chemical properties as characters—the "Acid Family" and the "Base Family"—students overcome the abstraction of molecular behaviors through clear visual cues, taste associations, and tactile textures.

The course bridges everyday experiences (such as dropping food on clothing during an *Aloo Paratha* lunch) with macro-scientific concepts (pH indicators, structural transformation, and safety protocols). Utilizing red cabbage extract ("Magic Juice") and turmeric water ("The Yellow Code"), students experience a live lab setting that stimulates observational reasoning, hypothesis formulation, and critical analysis.

> **Planrs Developmental Strategy:** Early childhood science must focus on perceptual shifts and tactile categorization. Rather than introducing terms like "hydrogen ion concentration" or "logarithmic scaling," this curriculum pairs sensory vocabulary (sour, bitter, slippery, sharp) directly with physical color transitions to construct permanent cognitive frameworks.

PEDAGOGICAL FRAMEWORK & OBJECTIVES

CURRICULUM DESIGN ARCHITECTURE

2. Learning Objectives & Competency Matrix

This unit maps directly to international primary science standards, focusing on physical properties of matter, chemical changes, and systematic inquiry.

Domain	Core Competency Objective	Success Indicator
Cognitive Taxonomy	Differentiate between acids and bases using sensory properties (taste, feel) and reaction colors.	Correctly identifies Lemon as an acid and Soap as a base during blind sorting.
Experimental Skill	Execute step-by-step indicator application using a dropping pipette with adult supervision.	Maintains fine-motor control to drop solution precisely into test wells.
Safety Awareness	Internalize and demonstrate the "No-Sip, No-Sniff" laboratory protocol.	Refrains from tasting unknown liquids and requests adult permission explicitly.
Analytical Reasoning	Connect real-world occurrences (stains changing color) to underlying chemical category shifts.	Explains why soap water turns an accidental turmeric stain rust-red.

Socio-Emotional Integration

By working through a structured mystery (The Mystery Stain), students develop collaborative problem-solving skills, learn to accept unexpected experimental outcomes, and build confidence in executing hands-on tasks. Safety rules are integrated not as restrictions, but as protective "Lab Detective Superpowers."

Key Curricular Focus: Fine Motor & Sensory Control
Using pipettes, handling small safety containers, and monitoring solution columns directly aids in bilateral coordination and visual tracking for younger learners.

LESSON PATHWAY & MILESTONES

60-90 MINUTE BREAKDOWN STRUCTURE

3. Session Timeline & Instructional Milestones

The session is divided into 5 clear chronological segments to maintain high engagement and accommodate short attention spans.

Phase	Time Allocation	Core Activity & Slide Focus	Teacher Priority
Phase 1: Hook	10 Mins	The Mystery Turmeric Stain from Lunch (Slides 2-3). Hook via sensory curiosity.	Maximize surprise; validate past student experiences with food stains.
Phase 2: Concept	15 Mins	Introducing Kingdom Acid & Kingdom Base (Slides 4-7). Characteristic mapping.	Reinforce the dual traits: Sour/Pink vs. Bitter-Slippery/Green-Blue.
Phase 3: Live Lab	25 Mins	The pH Rainbow Magic Color Show (Slide 8). Live indicator testing.	Ensure physical space organization; oversee individual student execution.
Phase 4: Synthesis	15 Mins	Detective Quiz & Cognitive Review (Slides 9-11). Multiple choice evaluation.	Formatively assess understanding; correct misconceptions on neutral substances.
Phase 5: Outro	10 Mins	Home Mission Briefing: The Yellow Code (Slides 12-14). Caregiver alignment.	Enforce safety warnings; hand out clear home lab worksheets.

Adaptive Extension Track

For advanced Grade 2 cohorts, the duration can be expanded to 90 minutes by introducing a third chemical category: **Neutrals** (Plain Water). Students will discover that some liquids are "peacekeepers" and do not trigger color changes in either indicator.

MATERIAL & REAGENT LOGISTICS

PREPARATION, QUANTITIES, AND SAFETY PROFILES

4. Live Lab Material Checklist & Setup Guide

All reagents must be prepared 30–45 minutes prior to the session start. All components are non-toxic, food-grade household items, aligning with strict early childhood guidelines.

1. The Indicators (The "Magic Juices")

- **Red Cabbage Indicator:** Chop 1 cup of red cabbage, steep in boiling water for 15 minutes, filter out solids, and collect the intense purple liquid. Cool to room temperature.
- **Turmeric Solution:** Dissolve 1 tablespoon of pure organic turmeric powder into 100ml of warm water. Agitate until thoroughly suspended.

2. Sample Solutions for Testing

- **Lucy Lemon Juice:** Freshly squeezed lemon juice, strained to eliminate pulp (High acidity).
- **Tamarind Water:** Softened tamarind pulp soaked in water, filtered (Medium acidity).
- **Sammy Soap Water:** Clear liquid dishwashing soap or hand soap diluted with warm water (Alkaline/Base).
- **Whitewash Water / Baking Soda Solution:** 1 tablespoon of sodium bicarbonate dissolved in 100ml water (Alkaline/Base).
- **Sour Curd Solution:** 1 tablespoon of fermented yogurt whisked into 50ml water (Test solution for Quiz).

Item Description	Quantity per Student	Safety Classification	Disposal Method
Red Cabbage Indicator (Purple)	30 mL	Completely Safe / Edible	Sink Drain
Turmeric Indicator (Yellow)	20 mL	Staining Hazard / Safe	Sink Drain
Diluted Soap Solution	15 mL	Eye Irritant (Mild)	Sink Drain
Baking Soda Solution	15 mL	Safe	Sink Drain
Plastic Dropping Pipettes (3ml)	3 Units	Mechanical Hazard Only	Wash & Reuse

LESSON EXECUTION: THE HOOK

SLIDE 2 & SLIDE 3 DETAILED BREAKDOWN

5. Detailed Scripting: The Mystery Aloo Paratha Stain

Visual Mapping & Presentation Focus

The session opens with Slide 2, displaying a bright illustration of a child discovering a vibrant yellow stain on their clothing. Slide 3 follows immediately, demonstrating the dramatic transition of that stain turning rust-red when wiped with soap. This acts as our primary anchor.

> ***Teacher Script (Verbatim):***
> *"Oh no, Little Detectives! Look at this slide! We just had a delicious lunch of warm, buttery Aloo Paratha. But look closely at the shirt... gasp! A bright yellow spot! Who can tell me what spice gives our food that beautiful yellow color? Yes, it's Turmeric! Now, watch what happens when we try to clean it up with some regular soap and water. *Click to Slide 3* Look! Wow! Did it disappear? No! It turned into a deep, magical secret RED color! How did a yellow stain turn red just by touching soap? Today, we are going to use our science glasses to solve this color mystery!"*

Pedagogical Strategy & Classroom Management

- **Engagement Check:** Have every child look at their own clothing or hold up a yellow coloring crayon to anchor the color concept.
- **Concept Bridge:** Do not give away the answer yet. Allow the students to guess if it is "magic," "paint," or "science."
- **Transition Cue:** Explain that to solve the stain mystery, we must travel to two invisible kingdoms that live right inside our kitchens.

> **CRITICAL SAFETY RULE ENFORCEMENT:**
> Introduce Slide 3's Safety Badge immediately. Teach the "No-Sip, No-Sniff" rule. Have students cross their arms and repeat: *"If it's in a lab space, we never sip, we never sniff, we always ask an adult!"*

LESSON EXECUTION: KINGDOM ACID

SLIDE 4 & SLIDE 5 DETAILED BREAKDOWN

6. Meet the Acid Family - The Sour Heroes

Character Profiles

Lucy Lemon: Bright, energetic, sharp, and intensely sour. Lives in citrus fruits.

Tamarind Water: Calm but tangy, deep-flavored, and carries a strong sour punch.

Core Scientific Attributes

- **Taste Category:** Sour ($pH < 7$).
- **Indicator Action:** Turns the dark purple Magic Juice into a stunning, bright pink-magenta!

Visual Indicators for Children:
ACID + Purple Cabbage Juice = **Bright Pink / Magenta Rose**

Children associate the sharp, face-puckering sensation of lemons with high energy and hot pink tones. This sensory grouping establishes an intuitive understanding of high hydrogen ion presence without needing formulas.

Teacher Script & Interaction

> ***Teacher Script (Verbatim):***
> *"Let's open the gates to Kingdom Acid! *Click to Slide 4* Meet the Acid Family! They are the ultimate Sour Flavor Heroes. When you bite into a lime or taste tangy tamarind, and your face goes like this—*make a squeezed sour face*—that's the Acid Family waving hello! They are sharp, fast, and they love bright colors. If you give them our purple Magic Juice, they cast a spell and turn it into a bright, flaming PINK-MAGENTA! Let's look at Slide 5—see that beautiful pink splash? That is Acid Magic!"*

Formative Assessment Check

Ask the class: *"If I give you an apple that tastes super sour, which family lives inside it?"* Expect a chorus of "The Acid Family!"

LESSON EXECUTION: KINGDOM BASE

SLIDE 6 & SLIDE 7 DETAILED BREAKDOWN

7. Meet the Base Family - The Smooth Friends

Character Profiles

Sammy Soap: Slippery, loves bubbles, smooth to touch, and tastes bitter (though we never eat him!).

Whitewash Water (Baking Soda): Calm, soft, looks chalky, and loves cleaning things up.

Core Scientific Attributes

- **Texture/Taste:** Slippery feel, bitter taste ($pH > 7$).
- **Indicator Action:** Turns our purple Magic Juice into a cool, refreshing blue or deep emerald green!

Visual Indicators for Children:
BASE + Purple Cabbage Juice = **Cool Blue** / **Emerald Green**

Bases are linked with soothing, cool tones like ocean blue and forest green. The physical sensation of slipperiness helps children categorize cleaners, detergents, and alkaline suspensions uniquely.

Teacher Script & Interaction

Teacher Script (Verbatim):
*"Now let’s cross the river to meet their neighbors. *Click to Slide 6* Meet the Base Family! They are completely different from the Acid Family. They aren't sour at all—they are smooth, slippery, and super calm. Think of soapy water when you wash your hands—how it slides smoothly over your skin. That’s Sammy Soap! The Base Family absolutely loves cool colors like ocean blue and green. When we add them to our purple Magic Juice, look at Slide 7—wow! It turns into an amazing green-blue sea color!"*

Guided Group Discussion

Have the children rub their hands together as if washing them with soap. Ask: *"What does that feel like? Is it sharp like a lemon, or smooth and slippery?"* Guide them to respond: "Slippery! It's a base!"

EXPERIMENTAL PHASE: LIVE LAB SHOW

SLIDE 8 PRACTICAL PROTOCOL GUIDE

8. Step-by-Step Practical Live Lab Instructions

This section outlines the workflow for Slide 8: **The pH Rainbow Live Lab**. Ensure all students have their safe sample cups lined up from left to right.

Lab Bench Alignment Layout:
Cup 1: Lemon Juice (Acid) | Cup 2: Pure Water (Neutral) | Cup 3: Soap Solution (Base)

Step-by-Step Procedure for the Classroom

1. **The Inspection:** Have students look at the central bowl of "Magic Juice" (Red Cabbage Extract). Confirm its native color: *Deep Purple*.
2. **The Acid Test:** Instruct students to fill their plastic pipette with Magic Juice. Carefully drop 5 drops into **Cup 1 (Lemon Juice)**. Swirl gently. Observe the rapid transition to vibrant pink.
3. **The Base Test:** Refill the pipette with Magic Juice. Drop 5 drops into **Cup 3 (Soap Solution)**. Swirl gently. Observe the transition to bright green or blue.
4. **The Control Test (Optional):** Drop 5 drops into **Cup 2 (Plain Water)**. Observe that the solution stays purple—no change!

Reaction Container	Starting Color	Added Reagent	Final Rainbow Result
Cup 1: Lemon Juice	Clear / Pale Yellow	Purple Magic Juice	**Bright Pink-Magenta**
Cup 2: Plain Water	Clear	Purple Magic Juice	**Deep Purple (No Change)**
Cup 3: Soap Solution	Clear / Opaque	Purple Magic Juice	**Emerald Green / Blue**

COGNITIVE SYNTHESIS & SPECTRUM SECRET

SLIDE 9 CONCEPTUAL INTEGRATION

9. Decoding the Rainbow's Secret

Slide 9 functions as the primary summary chart of the lesson, consolidating the chemical properties before entering the diagnostic evaluation phase. It summarizes the observations into a simple reference chart.

The Rainbow's Secret Code Breakdown

- **Red** / **Pink Outcomes:** Indicates interaction with **Sour Acids**. Examples discovered: Lemon juice, tamarind extract.
- **Green** / **Blue Outcomes:** Indicates interaction with **Smooth Bases**. Examples discovered: Soap concentrate, whitewash water, baking soda.
- **Purple Outcomes:** Indicates interaction with **Plain Neutral Water**. This shows that the substance is balanced, safe, and does not belong to either kingdom exclusively.

The Analytical Breakthrough Questions

Teachers should pose these structured questions to prompt early scientific deduction:

1. *"If our indicator fluid is naturally purple, why does it change color instead of just diluting?"*
   Target Logic: Because the hidden molecules inside the acid or base are actively altering the indicator's structure!
2. *"Can a liquid be both super sour and super slippery at the exact same time?"*
   Target Logic: No, they are opposite chemical families that neutralize each other when mixed.

Planrs Educational Insight: By teaching children that color is an expression of an internal state rather than an arbitrary trait, we prepare them for advanced atomic theory, molecular behavior, and analytical observations later on.

DIAGNOSTIC ASSESSMENT: CASE #1

SLIDE 10 DETECTIVE QUIZ DEEP DIVE

10. Formative Quiz Evaluation: The Sour Curd Case

Slide 10 challenges students to apply their newly built categorization framework to an unfamiliar substance: **Sour Curd (Yogurt)**.

> **Detective Problem Statement:**
> "Detective Elephant is holding a dropper filled with purple Magic Juice over a bowl of thick, creamy, sour curd. What color will the Magic Juice turn when it splashes into the curd?"

The Structural Evaluation Options

Option A: Bright Pink / Magenta Bowl
✔ **CORRECT ANSWER.** Curd contains lactic acid. The presentation emphasizes the keyword "SOUR," which links directly back to the Acid Family characteristic profile.

Option B: Emerald Green Ocean
✘ **INCORRECT.** Green indicates a smooth, slippery base. Curd is not basic or alkaline.

Option C: Plain Unchanged Purple
✘ **INCORRECT.** Purple indicates a neutral state like plain water. The curd is sour and will actively drive a color change.

Instructional Guide & Response Analysis

If students choose Option B, guide them by asking: *"When you taste plain curd, does it taste like soap or does it taste tangy and sharp like a lemon?"* This pivots their attention away from the visual appearance (white/creamy) and back to the core chemical indicator: **taste profile**.

DIAGNOSTIC ASSESSMENT: CASE #2

SLIDE 11 DETECTIVE QUIZ DEEP DIVE

11. Formative Quiz Evaluation: The Soap Water Test

Slide 11 switches focus to test the retention of the Base Family profile. This question tests if students can map properties backward from a named substance to its color output.

> **Detective Problem Statement:**
> "A lab assistant hands you a beaker filled with clear, slippery soap water. If you drop our purple cabbage Magic Juice into it, what color option from Slide 11 will appear?"

The Structural Evaluation Options

Option A: Bright Pink-Red Spheres
✘ **INCORRECT.** This would mean soap water is an acid, which contradicts its smooth, non-sour physical reality.

Option B: Cool Sea Green / Ocean Blue
✔ **CORRECT ANSWER.** Soap water is a traditional base. The anthocyanins in cabbage juice change structural confirmation at a high pH, turning blue-green.

Option C: Deep Neutral Purple
✘ **INCORRECT.** Soap alters the indicator state significantly; it will not remain neutral purple.

Classroom Feedback Loop Protocol

Have the students shout out the character name associated with this answer: **"Sammy Soap!"** to reinforce character-driven recall. Encourage them to verify this visual transformation during the upcoming home lab assignment.

THE TURMERIC WATER PUZZLE SOLVED

SLIDE 12 ANALYTICAL CONCLUSION

12. Solving the Mystery of the Aloo Paratha Stain

Slide 12 brings the lesson full circle by answering the opening mystery puzzle using their new scientific knowledge.

The Chemical Explanation:
Turmeric contains a compound called **curcumin**. Curcumin is a natural yellow indicator. It stays yellow when it touches neutral water or acids, but turns a bright rust-red when it touches a base like soap water!

Analytical Breakdown for Early Learners

Target Substance	Substance Category	Turmeric Color Output	Real-World Example
Lemon Juice / Vinegar	ACID	**Stays Bright Yellow**	No change on food stain
Pure Tap Water	NEUTRAL	**Stays Bright Yellow**	Washing with water alone
Detergent / Soap Solution	BASE	**Turns Deep Rust-Red**	Wiping with soap water

Teacher Script (Verbatim):

Teacher Script (Verbatim):
"Remember our opening Aloo Paratha stain? Now we have the scientific key to solve it! Turmeric is an indicator, just like our purple Magic Juice. It acts like a silent alarm. It loves staying bright yellow when it's alone or with acids. But the second Sammy Soap touches it—boom! The alarm goes off and it turns bright red! So when your yellow stain turned red, it was simply the turmeric yelling: 'Hey! I just found a base!'"

HOME MISSION: THE YELLOW CODE

SLIDE 13 & SLIDE 14 EXPERIENTIAL HOMEWORK

13. Structured Extension Activity: Caregiver-Guided Lab

To lock in what they learned, students are given an experiential assignment called **"The Yellow Code"**, as shown on Slides 13 and 14.

MANDATORY ADULT SUPERVISION PROTOCOL:
This home experiment requires a parent or adult guardian to handle the solutions. Students act as the "Lead Recording Detectives," taking notes on color changes without handling concentrated cleaning agents directly.

Home Lab Instructions for Families

1. **Prepare the Yellow Alarm Fluid:** Mix 1 teaspoon of turmeric powder into a small glass of warm water until it forms a uniform yellow liquid.
2. **Prepare the Testing Agents:** Set out two small transparent cups. Fill Cup A with white kitchen vinegar or lemon juice. Fill Cup B with concentrated laundry or hand soap dissolved in water.
3. **Execute the Color Transition:** Use a spoon or dropper to add 1 tablespoon of the yellow turmeric water into Cup A. Note the reaction. Next, add 1 tablespoon into Cup B. Watch the sudden change to rust-red.
4. **The Reset Experiment (Advanced):** Squeeze lemon juice directly into the red solution in Cup B. Watch as the high acid content neutralizes the base, turning the solution back to yellow!

Student Observation Log Sheet Structure

Experiment Step	Predicted Color	Observed Color	Family Identified
Turmeric + Lemon Juice	________________	________________	Acid Family 🟥
Turmeric + Soap Concentrate	________________	________________	Base Family 🟦

COURSE CONCLUSION & AUDIT METRICS

PLANRS ACADEMY EDUCATIONAL STANDARDS ASSURANCE

14. Post-Session Review & Teacher Metrics

As the session closes out on Slide 14 with a celebratory rainbow and star graphic, teachers must execute a quick post-course audit to track how well students understood the material.

Key Review Checkpoints

- Ensure every student has stopped handling their dropping pipettes and safely capped all reagent vials.
- Verify that students can explain why the turmeric food stain turned red when wiped with soap.
- Distribute copies of the "Yellow Code Home Lab Worksheet" along with parent safety alignment emails.

Curriculum Evaluation Benchmarks

Evaluation Target Metric	Excellent Level (80%+)	Developing Level (50-80%)	Requires Intervention (<50%)
Safety Rule Recall	Instantly repeats "No-Sip, No-Sniff" without prompting.	Remembers the rule only after seeing the slide badge.	Attempts to smell or taste materials during the lab.
Color Property Mapping	Correctly links Pink to Acid and Green/Blue to Base.	Mixes up colors but corrects themselves when prompted with character names.	Cannot match colors to character behaviors.
Real-World Application	Explains that soap water is a base that triggers the turmeric color change.	Identifies soap as a base but can't explain the reaction with turmeric.	Views the color change as a magic trick without scientific reasoning.

Curriculum Engineering Certification Statement:
The content within this Master Course Brief aligns with primary science standards for early childhood development. All narrative arcs, character designs, and material logistics have been reviewed for safety, accuracy, and clear lesson progression.

Primary Curriculum Source Document Reference:

PDF Resource: `PH Rainbow slides-compressed.pdf` (Slides 1 through 14 inclusive).